

Figure 5: Performance of the first task on Corafull (a), Amazon Computers (b), PPI (c) and Reddit (d), as more tasks are learned.

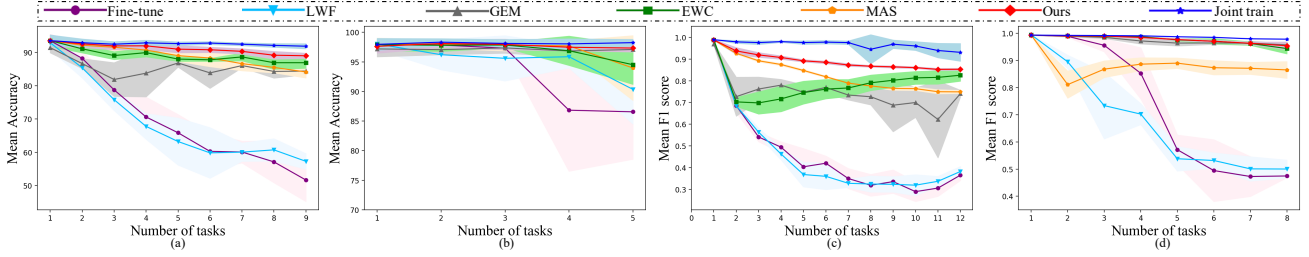


Figure 6: Average performance across all the learned tasks on Corafull (a), Amazon Computers (b), PPI (c), and Reddit (d) as more tasks are learned.

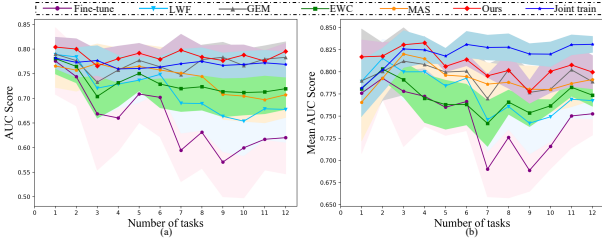


Figure 7: Performance of the first task (a) and average performance across all the learned tasks (b) as more tasks are learned on Tox21 dataset based on GATs.

Table 2: Performance comparison on the Tox21 dataset.

Methods	GATs		GCNs	
	AP ($\uparrow$)	AF ($\downarrow$)	AP ($\uparrow$)	AF ($\downarrow$)
Fine-tune	0.753 $\pm$ 0.024	0.084 $\pm$ 0.021	0.737 $\pm$ 0.028	0.098 $\pm$ 0.011
LWF	0.768 $\pm$ 0.041	0.058 $\pm$ 0.043	0.769 $\pm$ 0.034	0.059 $\pm$ 0.042
GEM	0.789 $\pm$ 0.010	0.036 $\pm$ 0.017	0.768 $\pm$ 0.041	0.037 $\pm$ 0.006
EWC	0.774 $\pm$ 0.025	0.038 $\pm$ 0.027	0.803 $\pm$ 0.031	0.014 $\pm$ 0.015
MAS	0.790 $\pm$ 0.023	0.032 $\pm$ 0.024	0.805 $\pm$ 0.021	0.014 $\pm$ 0.005
Ours	0.801 $\pm$ 0.019	0.022 $\pm$ 0.012	0.810 $\pm$ 0.024	0.013 $\pm$ 0.012
Joint train	0.831 $\pm$ 0.009	0.006 $\pm$ 0.009	0.822 $\pm$ 0.017	0.012 $\pm$ 0.008

function. Next, we verify the effectiveness of the topological information of graphs in graph-based continual learning by conducting experiments with the proposed TWP module. According to Table 3, TWP achieves consistent better performance on these two datasets, showing the effectiveness of preserving topological structure information to remember old tasks. Finally, we promote minimized importance scores for all parameters to preserve model capacity for future tasks, which further improves overall performance.

Table 3: Ablation study with GATs as the base model. W/_Loss and W/_TWP measures importance of parameters based on task-related loss and the proposed TWP module respectively. Full is the proposed method with all parts.

Methods	Corafull		Amazon Computers	
	AP ($\uparrow$)	AF ($\downarrow$)	AP ($\uparrow$)	AF ($\downarrow$)
W/_Loss	86.9 $\pm$ 1.7%	6.4 $\pm$ 1.8%	94.5 $\pm$ 3.3%	4.6 $\pm$ 4.5%
W/_TWP	88.6 $\pm$ 1.0%	3.7 $\pm$ 2.0%	96.2 $\pm$ 2.2%	0.8 $\pm$ 1.1%
Full	89.0 $\pm$ 0.8%	3.3 $\pm$ 0.3%	97.3 $\pm$ 0.6%	0.6 $\pm$ 0.2%

Conclusion

In this paper, we propose a dedicated continual learning method for graph neural networks, which is to our best knowledge the first attempt along this line. Specifically, we design a topology-aware weight preserving module which explicitly captures the topological information of graphs and measures the importance of the network’s parameters based on the task-related loss function and the topological information. When learning a new task, changes to the important parameters will be penalized to remember old tasks. Moreover, the proposed approach can be readily extended to arbitrary GNNs. The extensive experiments on both node-level tasks and graph-level one demonstrates the effectiveness and applicability of the proposed continual learning method on the graph domain.

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